Project: Worked examples
Title: Member checks (EN 1993-1-1)
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1. Scope and method

Calculates the resistance of a class 1, 2 or 3 steel member to tension, compression, buckling, bending, shear and their interaction, to EN 1993-1-1. Tension is positive; class 4 sections and lateral-torsional buckling are outside this sheet.

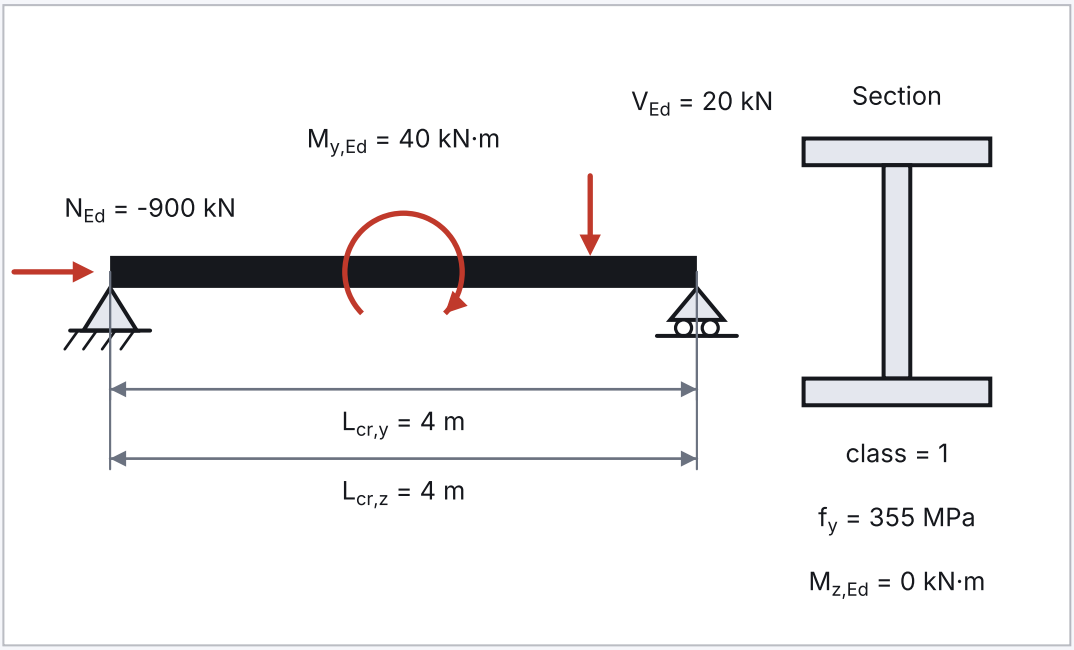


Figure 1 — A steel member under axial force, bending and shear, its buckling lengths about both axes, and its section. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Axial plastic resistance	$N_{pl,Rd} = A\, f_y/\gamma_{M0}$
Bending resistance, plastic modulus for class 1–2 and elastic for class 3	$M_{c,Rd} = W\, f_y/\gamma_{M0}$
Shear resistance	$V_{pl,Rd} = A_v\, f_y/(\sqrt{3}\,\gamma_{M0})$
Bending resistance reduced by high shear, when $V_{Ed} > 0.5\, V_{pl,Rd}$	$\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$. Class 1–2 I-section with h, t_w, t_f entered: $M_{y,V,Rd} = (W_{pl,y} - \rho\, A_w^2/(4\, t_w))\, f_y/\gamma_{M0}$, $A_w = (h - 2t_f)\, t_w$, and $M_{z,c,Rd}$ is not reduced; otherwise the whole resistance × $(1 - \rho)$
Elastic critical load and non-dimensional slenderness	$N_{cr} = \pi^2 EI/L_{cr}^2$; $\bar{\lambda} = \sqrt{(A\, f_y/N_{cr})}$
Buckling reduction factor and buckling resistance	$\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$; $N_{b,Rd} = \chi\, A\, f_y/\gamma_{M1}$
Cross-section check, linear interaction	$N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$

WHAT IT CALCULATES	RELATION
Member check, axial force and bending with buckling (6.61, 6.62)	$N_{Ed}/N_{b,Rd,min} + k_{yy}\, M_{y,Ed}/(W_y\, f_y/\gamma_{M1}) + k_{zz}\, M_{z,Ed}/(W_z\, f_y/\gamma_{M1}) \le 1$, the characteristic bending resistance M_{Rk} over γ_{M1} without the shear reduction

Basis: Tension[1](cl. 6.2.3); Compression[1](cl. 6.2.4); Bending moment[1](cl. 6.2.5); Shear[1](cl. 6.2.6); Bending and shear[1](cl. 6.2.8); Linear summation of utilisations[1](cl. 6.2.1(7)); Uniform members in compression: buckling resistance, buckling curves[1](cl. 6.3.1.1, 6.3.1.2); Uniform members in bending and axial compression[1](cl. 6.3.3); Imperfection factors for the buckling curves[2](§6.3.1, 1st ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$section_{class}$	Cross-section class	1		assumption: UC 203×203×46 S355 is class 1 in compression (Blue Book).	any
$curve_y$	Buckling curve about y-y	b		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, y-y	any
$curve_z$	Buckling curve about z-z	c		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, z-z	any
ltb	Compression flange against lateral-torsional buckling (optional; blank = blocks when $M_{y,Ed}$ is above 0)	Restrained		assumption: The column is held at both ends by the floor structure, which restrains its compression flange against lateral-torsional buckling (6.3.2 is outside this sheet).	any
A	Area	58.7	cm²	reference: S1 section sheet / Blue Book UC 203×203×46	1 … 100000 cm²
$W_{pl,y}$	Plastic section modulus about y-y	497	cm³	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm³
$W_{el,y}$	Elastic section modulus about y-y	450	cm³	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm³
$W_{pl,z}$	Plastic section modulus about z-z	231	cm³	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm³
$W_{el,z}$	Elastic section modulus about z-z	152	cm³	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm³
I_y	Second moment of area about y-y	4568	cm⁴	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm⁴

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
I_z	Second moment of area about z-z	1548	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
h	Depth for the shear area (optional; blank = use A_v)	203.2	mm	spec: BS EN 10365 UC 203×203×46	1 … 5000 mm
b	Width for the shear area (optional; blank = use A_v)	203.6	mm	spec: BS EN 10365 UC 203×203×46	1 … 5000 mm
t_w	Web thickness for the shear area (optional; blank = use A_v)	7.2	mm	spec: BS EN 10365 UC 203×203×46	0.1 … 500 mm
t_f	Flange thickness for the shear area (optional; blank = use A_v)	11	mm	spec: BS EN 10365 UC 203×203×46	0.1 … 500 mm
r	Root radius for the shear area (optional; blank = 0)	10.2	mm	spec: BS EN 10365 UC 203×203×46	0 … 500 mm
f_y	Yield strength for the thickness	355	MPa	spec: BS EN 10025-2 S355, t ≤ 16 mm	200 … 700 MPa
E	Young's modulus	210	GPa	spec: BS EN 1993-1-1 3.2.6	190 … 220 GPa
N_{Ed}	Design axial force, tension positive and compression negative, opposite in sign to the base plate sheet's axial force	-900	kN	reference: S2 frame sheet, column 1, ULS	-100000 … 100000 kN
$M_{y,Ed}$	Design moment about y-y (major axis)	40	kN·m	reference: S2 frame sheet, column 1, ULS	0 … 100000 kN·m
$M_{z,Ed}$	Design moment about z-z (minor axis)	0	kN·m	reference: S2 frame sheet	0 … 100000 kN·m
V_{Ed}	Design shear force	20	kN	reference: S2 frame sheet	0 … 100000 kN
$L_{cr,y}$	Buckling length about y-y	4	m	assumption: Pinned both ends: L_cr = L = 4 m.	0.1 … 100 m
$L_{cr,z}$	Buckling length about z-z	4	m	assumption: Pinned both ends: L_cr = L = 4 m.	0.1 … 100 m
γ_{M0}	Partial factor for cross-section resistance	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
γ_{M1}	Partial factor for member buckling	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
γ_{M2}	Partial factor for net-section fracture	1.1		default	1 … 1.35
η	Shear-area factor	1		spec: UK NA to BS EN 1993-1-5 NA.2.4	1 … 1.2
k_{yy}	Interaction factor for bending about y-y in the member check	1		assumption: Simplified interaction: k_yy = 1.0.	0.5 … 2
k_{zz}	Interaction factor for bending about z-z in the member check	1		assumption: Simplified interaction: k_zz = 1.0.	0.5 … 2

3. Assumptions

A1. The cross-section class is the class the user states, 1, 2 or 3. (Class 4 effective properties are not derived here.)
A2. The compression flange is restrained against lateral-torsional buckling, as the input ltb says. (Lateral-torsional buckling of a member that is not restrained is checked on the LTB interaction sheet.)
A3. A k left at 1.0 is replaced by the Annex A or B upper bound $C_m(1 + 0.8n)$. (Applies under compression with a moment; the smaller flexural-buckling reduction is used on the axial term and the factors are not derived.)
A4. Imperfection factors are the textbook values for the buckling curve unless the user overrides them. (Gardner and Nethercot, Designers' Guide to EN 1993-1-1.)
A5. Partial factors are 1.0 as in the UK National Annex unless the user overrides them. (UK National Annex to BS EN 1993-1-1.)
A6. $\text{section}_{\text{class}}$ (Cross-section class) is taken as 1. (UC 203×203×46 S355 is class 1 in compression (Blue Book).)
A7. ltb (Compression flange against lateral-torsional buckling (optional; blank = blocks when $M_{y,Ed}$ is above 0)) is taken as Restrained. (The column is held at both ends by the floor structure, which restrains its compression flange against lateral-torsional buckling (6.3.2 is outside this sheet).)
A8. $L_{cr,y}$ (Buckling length about y-y) is taken as 4 m. (Pinned both ends: L_{cr} = L = 4 m.)
A9. $L_{cr,z}$ (Buckling length about z-z) is taken as 4 m. (Pinned both ends: L_{cr} = L = 4 m.)
A10. k_{yy} (Interaction factor for bending about y-y in the member check) is taken as 1. (Simplified interaction: k_{yy} = 1.0.)
A11. k_{zz} (Interaction factor for bending about z-z in the member check) is taken as 1. (Simplified interaction: k_{zz} = 1.0.)
A12. The compression flange is restrained against lateral-torsional buckling, as the input ltb says. (Input ltb.)

4. Working

[28] $r_h = r$
$= 10.2\ \text{mm}$
$= 10.2\ \text{mm}$
Root radius used by the helper

[29] $A_{v,\text{rolled}} = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f$ ³
$= 58.7\ \text{cm}^2 - 2 \times 203.6\ \text{mm} \times 11\ \text{mm} + (7.2\ \text{mm} + 2 \times 10.2\ \text{mm}) \times 11\ \text{mm}$
$= 16.944\ \text{cm}^2$
Rolled I shear area, load parallel to the web

[30] $A_{v,\text{min}} = \eta * (h - 2 * t_f) * t_w$ ³
$= 1 \times (203.2\ \text{mm} - 2 \times 11\ \text{mm}) \times 7.2\ \text{mm}$
$= 13.046\ \text{cm}^2$
Lower bound η h_w t_w

[31] $A_v = \max(A_{v,rolled}, A_{v,min})$
$= \max(16.944\ cm^2, 13.046\ cm^2)$
$= 16.944\ cm^2$
Shear area
[32] $N_{pl,Rd} = A * f_y / \gamma_{M0}$ [4](6.2.4(2))
$= 58.7\ cm^2 \times 355\ MPa / 1$
$= 2083.8\ kN$
Plastic axial resistance
[33] $W_y = W_{pl,y}$ [5]
$= 497\ cm^3$
$= 497\ cm^3$
Section modulus about y (plastic, class 1–2)
[34] $W_z = W_{pl,z}$ [5]
$= 231\ cm^3$
$= 231\ cm^3$
Section modulus about z (plastic)
[35] $V_{pl,Rd} = A_v * f_y / (\sqrt{3} * \gamma_{M0})$ [6]
$= 16.944\ cm^2 \times 355\ MPa / (\sqrt{3} \times 1)$
$= 347.28\ kN$
Plastic shear resistance
[36] $u_V = V_{Ed} / V_{pl,Rd}$
$= 20\ kN / 347.28\ kN$
$= 0.05759$
Shear utilisation
[37] $\rho_{shear} = 0$ [7]
$= 0$
Shear reduction factor ($V_{Ed} \le 0.5\ V_{pl,Rd}$)
[38] $M_{y,c,Rd} = (1 - \rho_{shear}) * W_y * f_y / \gamma_{M0}$ [8](6.2.8)
$= (1 - 0) \times 497\ cm^3 \times 355\ MPa / 1$
$= 176.44\ kN{\cdot}m$
Bending resistance about y
[39] $M_{z,c,Rd} = (1 - \rho_{shear}) * W_z * f_y / \gamma_{M0}$ [5]
$= (1 - 0) \times 231\ cm^3 \times 355\ MPa / 1$
$= 82.005\ kN{\cdot}m$
Bending resistance about z
[40] $u_{M,y} = M_{y,Ed} / M_{y,c,Rd}$
$= 40\ kN{\cdot}m / 176.44\ kN{\cdot}m$
$= 0.22671$
Bending utilisation, y
[41] $u_{M,z} = M_{z,Ed} / M_{z,c,Rd}$
$= 0\ kN{\cdot}m / 82.005\ kN{\cdot}m$
$= 0$
Bending utilisation, z
[42] $N_{abs} = \text{abs}(N_{Ed})$
$= \text{abs}(-900\ kN)$
$= 900\ kN$
Axial force magnitude
[43] $u_N = N_{abs} / N_{pl,Rd}$
$= 900\ kN / 2083.8\ kN$
$= 0.43189$
Compression (cross-section) utilisation
[44] $u_{section} = u_N + u_{M,y} + u_{M,z}$ [9]
$= 0.43189 + 0.22671 + 0$
$= 0.65861$
Linear cross-section interaction

[47] α_y = imperfection factor, curve b [10]
= Designers' Guide §6.3.1
= 0.34
Imperfection factor about y
[48] α_z = imperfection factor, curve c [10]
= Designers' Guide §6.3.1
= 0.49
Imperfection factor about z
[49] $N_{cr,y}$ = pi^2 * E * I_y / $L_{cr,y}$^2 [11]
= π^2 × 210 GPa × 4568 cm⁴ / (4 m)^2
= 5917.3 kN
Elastic critical force about y
[50] λ_y = sqrt(A * f_y / $N_{cr,y}$) [11]
= sqrt(58.7 cm² × 355 MPa / 5917.3 kN)
= 0.59343
Non-dimensional slenderness about y
[51] Φ_y = 0.5 * (1 + α_y * (λ_y - 0.2) + λ_y^2) [11]
= 0.5 × (1 + 0.34 × (0.59343 - 0.2) + 0.59343^2)
= 0.74296
Φ about y
[52] χ_y = min(1, 1 / (Φ_y + sqrt(Φ_y^2 - λ_y^2))) [11]
= min(1, 1 / (0.74296 + sqrt(0.74296^2 - 0.59343^2)))
= 0.84034
Reduction factor χ about y
[53] $N_{b,y,Rd}$ = χ_y * A * f_y / γ_{M1} [12]
= 0.84034 × 58.7 cm² × 355 MPa / 1
= 1751.1 kN
Buckling resistance about y
[54] $N_{cr,z}$ = pi^2 * E * I_z / $L_{cr,z}$^2 [11]
= π^2 × 210 GPa × 1548 cm⁴ / (4 m)^2
= 2005.3 kN
Elastic critical force about z
[55] λ_z = sqrt(A * f_y / $N_{cr,z}$) [11]
= sqrt(58.7 cm² × 355 MPa / 2005.3 kN)
= 1.0194
Non-dimensional slenderness about z
[56] Φ_z = 0.5 * (1 + α_z * (λ_z - 0.2) + λ_z^2) [11]
= 0.5 × (1 + 0.49 × (1.0194 - 0.2) + 1.0194^2)
= 1.2204
Φ about z
[57] χ_z = min(1, 1 / (Φ_z + sqrt(Φ_z^2 - λ_z^2))) [11]
= min(1, 1 / (1.2204 + sqrt(1.2204^2 - 1.0194^2)))
= 0.52876
Reduction factor χ about z
[58] $N_{b,z,Rd}$ = χ_z * A * f_y / γ_{M1} [12]
= 0.52876 × 58.7 cm² × 355 MPa / 1
= 1101.9 kN
Buckling resistance about z
[59] $N_{b,Rd}$ = min($N_{b,y,Rd}$, $N_{b,z,Rd}$)
= min(1751.1 kN, 1101.9 kN)
= 1101.9 kN
Governing buckling resistance

[60] $u_{buckling} = N_{abs} / N_{b,Rd}$ [13]
$= 900\ kN / 1101.9\ kN$
$= 0.8168$
Flexural buckling utilisation
[61] $M_{y,b,Rd} = W_y * f_y / \gamma_{M1}$ [14](6.61)
$= 497\ cm^3 \times 355\ MPa / 1$
$= 176.44\ kN \cdot m$
Characteristic bending resistance about y over γ_{M1} (member check, no shear reduction)
[62] $M_{z,b,Rd} = W_z * f_y / \gamma_{M1}$ [14](6.62)
$= 231\ cm^3 \times 355\ MPa / 1$
$= 82.005\ kN \cdot m$
Characteristic bending resistance about z over γ_{M1} (member check, no shear reduction)
[63] $u_{member} = N_{abs} / N_{b,Rd} + k_{yy} * M_{y,Ed} / M_{y,b,Rd} + k_{zz} * M_{z,Ed} / M_{z,b,Rd}$ [15]
$= 900\ kN / 1101.9\ kN + 1 \times 40\ kN \cdot m / 176.44\ kN \cdot m + 1 \times 0\ kN \cdot m / 82.005\ kN \cdot m$
$= 1.0435$
Simplified member interaction
[64] $u_{limit} = limit$
$= the\ utilisation\ limit$
$= 1$
Utilisation limit

5. Results

SYMBOL	VALUE	DESCRIPTION
$N_{pl,Rd}$	2083.8 kN	Plastic axial resistance
$N_{b,Rd}$	1101.9 kN	Governing buckling resistance
$M_{y,c,Rd}$	176.44 kN·m	Bending resistance about y-y
$M_{z,c,Rd}$	82.01 kN·m	Bending resistance about z-z
$V_{pl,Rd}$	347.28 kN	Plastic shear resistance
ρ_{shear}	0	Shear reduction factor ρ
$u_{section}$	0.66	Cross-section interaction
$u_{buckling}$	0.82	Buckling utilisation
u_{member}	1.05	Member interaction (governing)
A_v	16.94 cm²	Shear area
$M_{y,b,Rd}$	176.44 kN·m	Characteristic bending resistance about y-y over the partial factor
$M_{z,b,Rd}$	82.01 kN·m	Characteristic bending resistance about z-z over the partial factor
$N_{cr,y}$	5917.3 kN	Elastic critical force about y-y
$N_{cr,z}$	2005.3 kN	Elastic critical force about z-z
λ_y	0.59343	Non-dimensional slenderness about y-y
λ_z	1.0194	Non-dimensional slenderness about z-z
χ_y	0.84034	Buckling reduction factor about y-y
χ_z	0.52876	Buckling reduction factor about z-z

SYMBOL	VALUE	DESCRIPTION
$N_{b,y,Rd}$	1751.1 kN	Buckling resistance about y-y
$N_{b,z,Rd}$	1101.9 kN	Buckling resistance about z-z
u_N	0.43	Axial utilisation
$u_{M,y}$	0.23	Bending y-y utilisation
$u_{M,z}$	0.00	Bending z-z utilisation
u_V	0.06	Shear utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Member interaction against its limit	1.0436	1	1.05	[16](6.3.3)
PASS	Axial force against buckling resistance	900 kN	1101.9 kN	0.82	[16](6.3.1.1)
PASS	Cross-section interaction against its limit	0.65861	1	0.66	[16](6.2.1(7))
PASS	Compression against plastic resistance	900 kN	2083.8 kN	0.43	[17](6.2.4)
PASS	Bending moment y-y against bending resistance	40 kN·m	176.44 kN·m	0.23	[16](6.2.5)
PASS	Shear force against plastic shear resistance	20 kN	347.28 kN	0.06	[16](6.2.6)
PASS	Bending moment z-z against bending resistance	0 kN·m	82.01 kN·m	0.00	[16](6.2.5)

7. Applicability

WARNING k_{yy}: entered as 1.0, but Annex A or B give k up to 1.411 here. With it the interaction would be 1.137 against 1.044 shown. [k-default]

Defaults used (inputs left empty):
γ_{M2}: not entered, so the default of 1.1 is used.

8. Verification

```
Template struct.en1993-member@1.6.0 Content hash 09dec90614ac2f64 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.

2. Gardner L, Nethercot DA. Designers' Guide to EN 1993-1-1: Eurocode 3: Design of Steel Structures, General Rules and Rules for Buildings. Thomas Telford; 2005.
3. 6.2.6(3).
4. 6.2.3(2),.
5. 6.2.5(2).
6. 6.2.6(2).
7. 6.2.8(2).
8. 6.2.5(2),.
9. 6.2.1(7).
10. 6.3.1.2.
11. 6.3.1.2(1).
12. 6.3.1.1(3).
13. 6.3.1.1(1).
14. 6.3.3, Eq.
15. 6.3.3(4).
16. cl.
17. cl. 6.2.3,.